Kumar Abhinav

AI Software Engineer

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Core Technologies & Skills

Languages & Frameworks:
Python · Java · TypeScript · FastAPI · Spring Boot · Angular

APIs & Integrations: RESTful APIs · Temporal Workflows · JIRA API · SAP

Data & Analytics:
SQL(Advanced) · Elasticsearch · Kibana · AWS Redshift · Pandas · Tableau

AI & Automation: OpenAI API, LangGraph Transformers, Agentic AI

AI Software Engineer with 4+ years building production-scale GenAI applications (GPT-3.5-turbo, GPT-4, Whisper). Expert in agentic-AI pipelines (LangGraph, Pinecone), Python/FastAPI microservices, and Temporal orchestration—achieving 50% query-latency reduction, 65% user adoption in 2 months, and 2,000+ engineer-hours saved annually while maintaining 99.9% uptime.

Blends technical depth with 12+ years of data & supply chain analytics expertise—driving measurable efficiency, adoption, and cost savings.

Career Highlights

- **Gen AI Application:** Built and deployed scalable Agentic AI solutions using LangGraph, Pinecone, GPT-4, **reducing triage time by 45%** and achieving 65% adoption
- **Small Language Model from Scratch:** Built a GPT-style transformer (15.2 M parameters) in PyTorch from scratch, achieving 97% accuracy on domain text generation with an optimized training pipeline and interactive generation interface.
- **Product Development:** Spearheaded the end-to-end design and launch of four bespoke analytics and automation products for healthcare & pharma clients—escalation-management APIs, real-time inventory & order sync API, country-level Inventory Lifecycle Dashboard, these solutions drove 25–40% efficiency gains, generated approx. 1 M USD in upsells, and achieved 65% user adoption.

Work Experience

LinkedIn | Sr. Software Engineer (Contractor), Sunnyvale, CA. Oct 2024 - Present

Own and ship full-stack features for LinkedIn's internal Employee Productivity Platform, serving 28K+ employees. Partner with Product, UX, and IT Service Management (ITSM) to define roadmaps, build APIs, and iterate on user feedback—delivering measurable impact.

Specific Contributions:

- **Product Development – Employee Productivity Platform:** Led end-to-end development of an internal **GenAI Q&A chatbot**, from requirements gathering to **Azure DevOps deployment** and UAT testing. Leveraged **Azure OpenAI Service** (GPT-4), GPT-4o, and **RAG (Azure AI Search)** to automate ticket resolution—**driving 65% adoption** in 2 months and **reducing MTTR by 45%.** Migrated Java legacy backend into Python/FastAPI, decommissioning entire java VMs and cutting platform complexity.
- **Product Features – ServiceNow, Contract & Finance Integrations: Integrated ServiceNow incident management** into chatbot, **reducing** unnecessary **ticket creation by 30%.** Delivered Contract Finder **(Salesforce + Microsoft Dynamics,** saving ~$50K in revenue leakage**) and Franklin Finance** integration (real-time PO queries). Partnered with Product team for UAT and rollout.
- **Data Pipeline & Engineering Excellence: Migrated employee data pipelines—**User Refresh (LDAP + Slack), User Priming (Cosmos), and Working Hours (MS Graph)—to **Temporal workflows, eliminating 95% of failures and reclaiming 20% of weekly engineering hours.** Modernized CI/CD pipelines with dependency cleanup and automated monitoring, **improving build times by 40%.**
- **Platform Engineering – Ticket Classification Agent:** Built internal classification agent to auto-tag ITSM tickets by category, urgency, and resolution type. Leveraged model 4.1 for context enrichment, cutting triage time by 35% and boosting **assignment accuracy to 92%. LLMOps & Evaluation**: Implemented prompt-level latency tracking, confusion-metric dashboards, and automatic retrieval scoring for RAG pipelines **(Precision@k, MRR, Recall@k).**

RFXCEL | Software Engineer, San Ramon, CA. May 2023 – Dec 2024

Partnered with Sales Engineering to architect and ship new products features, analytics and automation products for healthcare & pharma clients like cardinal, AbbVie and Walgreens etc. to driving efficiency, scalability, and user adoption.

Specific Contributions:

- **Internal Productivity:** Leveraged **GenAI (LLM Models GPT -4o)** to prototype and launch several tools for internal productivity automation such as **Text2SQL Query Assistant**: Built a GenAI-powered tool using **OpenAI's API (GPT -4o) and AWS Redshift** to translate **natural-language prompts into SQL queries** for Customer Support Analysts. Reduced support-ticket **processing time by 45%,** reduced the dependency to hire three contractor roles, and **saved $ 100 K /annually. Automated Release-Notes Generator:** Developed a **pipeline leveraging the JIRA API**, Confluence data, and OpenAI to draft customer-friendly release notes automatically at the end of each sprint. **Saved 576 Engineering Working hours / year** and **increased customer satisfaction by 65%.**

- **Jira Conversational Assistant (agentic AI) — 10× faster triage**: Built a LangGraph-based assistant that turns natural-language queries into ticket actions and escalation-risk scoring. **Triage dropped from ~30–40 min to <3–4 min per incident** (order-of-magnitude), **preventing ~20 escalations/month** and **saving ~16 engineer-hours/week**, and saved **~16 engineer-hours per week (~832 hours/year),** cutting **support costs by ~$120 K annually.**
- **Product Development – Real-Time Inventory Lifecycle Dashboard:** Built a SQL⟶ Elasticsearch pipeline feeding an Elasticsearch index and Kibana dashboard to deliver on-demand, country-level inventory and supply-chain insights. Reduced **data-query latency by 50%, boosted** operational efficiency **by 25%,** upsold the existing traceability platform to pharma clients, and **generated $ 1 M in additional revenue**. Empowered government regulators, suppliers, and manufacturers with actionable, real-time data
- **Product Development: Legacy product: two features: (a) Escalation management:** Developed Java Spring Boot REST APIs and an Angular UI to enable Walgreens and its distributors to configure exception rules in real time. Empowered users to define alerts on the fly—cutting **resolution time by 40%** and **reducing operational losses** (support staffing and logistics) **by approx. $ 1 M / year**. **(b) Automated real-time inventory & order synchronization:** Developed a Java Spring Boot public API to integrate RfXcel's platform with Manufacturer (AbbVie) SAP system—**automating 100% of order handoffs**, eliminating manual, error-prone updates, ensuring 100% accurate, up-to-date inventory and order status, and **saving $ 200 K / year in labor costs**.

Early Career | Business Data Analyst, Chennai/Mumbai, India. Apr 2009 – Aug 2022

Owned and delivered end-to-end supply-chain optimization products, combining SAP automation, advanced analytics, and forecasting to drive compliance, cost savings, and operational efficiency—culminating in successful McKinsey audits and $1 million in working capital improvement.

Specific projects:

- **Global Inventory Optimization Engine:** Designed and built a quantitative segmentation framework (JIT, VMI) integrated into SAP —leveraged Python and SQL to map equipment to spare parts across plants, enabling accurate inventory classification **reduced global inventory by 25%, freed up $1 M in working capital.**
- **Manufacturing Analytics & Loss-Reduction Module**: Developed a Python-based predictive analytics product leveraging linear regression machine learning model, hypothesis testing, and variance analysis to pinpoint production inefficiencies. Presented Executive **Tableau dashboard** for the leadership group. **Eliminated 22%** of manufacturing losses, and **reduced manufacturing cost from $200/MT to $150/MT.**
- **Operational Cost Modeling:** Conducted in-depth cost studies across four production units; built BOQ-driven financial models and "what-if" scenarios that **uncovered 8% in annual cost savings (≈$8 M).**

Skills

Functional: Distributed Systems, API Development, Performance Optimization, Asynchronous Programming, Process Mapping, Delivery Presentation, Agile, SDLC, Jira, Git, Temporal, CI/CD, Docker.

Personal Projects

Small Language Model from Scratch — Architected and trained a **GPT-style transformer (15.2M parameters)** in PyTorch from scratch. Implemented complete **training pipeline on Apple M2 GPU** with custom dataset preprocessing and batching. **Achieved 97% accuracy (perplexity: 1.03)** on domain-specific text generation task. Built interactive generation interface with temperature and top-k sampling controls. Optimized training with gradient accumulation, mixed precision, and thermal management for consumer hardware.

Education and Academic Background

M.S. in Business Analytics, Aug 2022 to Dec 2023 || **California State University East Bay**, Hayward, CA

B.Tech. in Chemical Engineering (Major) | **Computer Science & Engineering (Elective),** Aug 2005 to Dec 2009 || **Sathyabama Deemed University** Chennai, India

Portfolio – https://abhinav0905.github.io